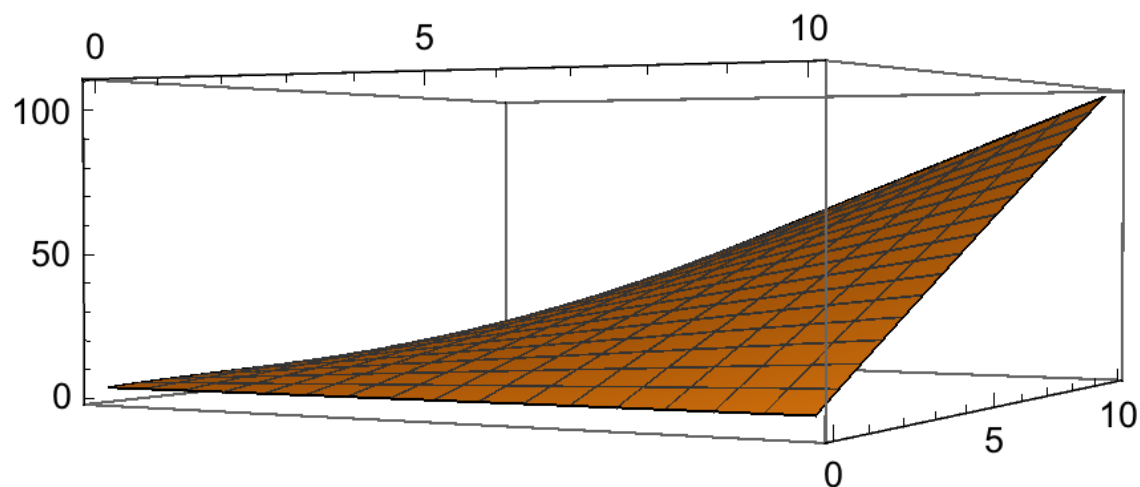


Multivariate Calculus

For our last two classes we'll deal with functions of more than one variable

Ex: A person's utility is a function of apples (x) and bananas (y), and looks like

$$U(x, y) = (5 + x)^{\frac{1}{2}} + (2y + 1)^{\frac{1}{2}} + xy$$



We want to see how U changes as x and y change

First, we'll consider what happens to utility as x changes and y remains fixed (we will treat y as a constant)

$\frac{\partial f}{\partial x}$ is the *partial derivative* of f with respect to x (with $\frac{\partial f}{\partial y}$ defined similarly)

$$U(x, y) = (5 + x)^{\frac{1}{2}} + (2y + 1)^{\frac{1}{2}} + xy$$

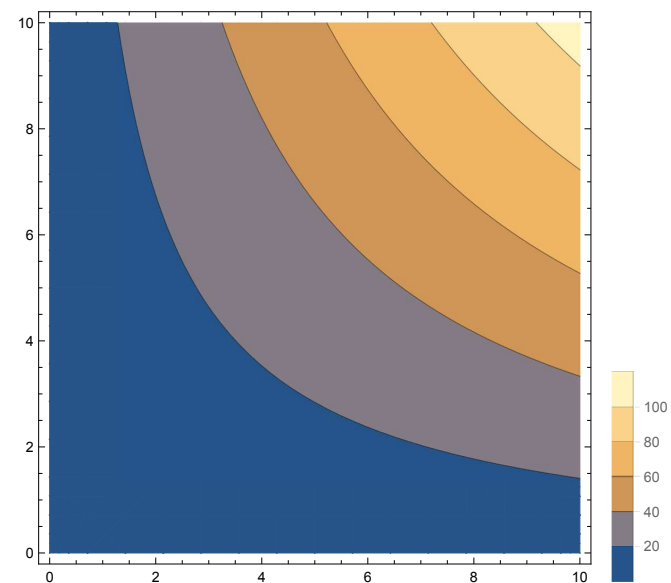
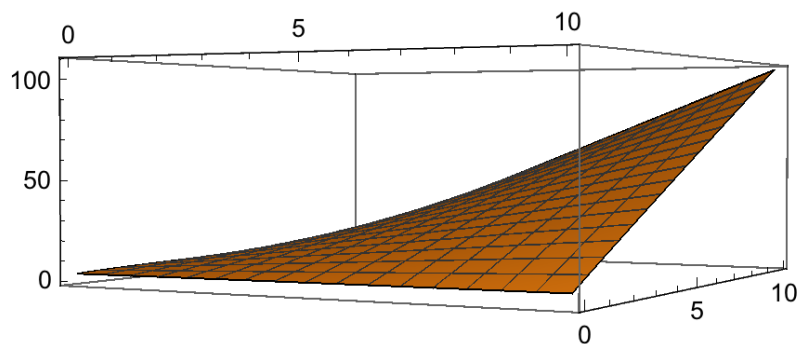
$$\frac{\partial U}{\partial x} = y + \frac{1}{2}(5 + x)^{-\frac{1}{2}}$$

$$\frac{\partial U}{\partial y} = x + \frac{1}{2}(2y + 1)^{-\frac{1}{2}} * 2$$

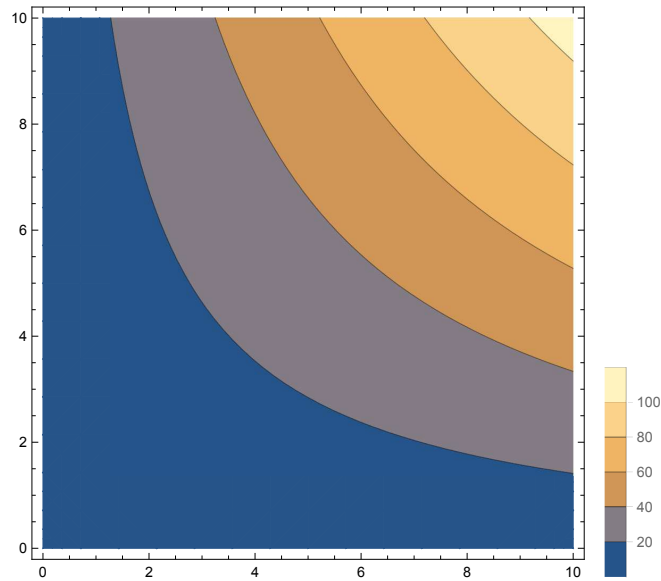
$$= x + (2y + 1)^{-\frac{1}{2}}$$

$$U(x, y) = (5 + x)^{\frac{1}{2}} + (2y + 1)^{\frac{1}{2}} + xy$$

$$\frac{\partial U}{\partial x} = y + \frac{1}{2}(5 + x)^{-\frac{1}{2}}$$



- $\frac{\partial U}{\partial x}$ always positive
- $\frac{\partial U}{\partial x}$ greater for greater y
- $\frac{\partial U}{\partial x}$ smaller for greater x



- $\frac{\partial U}{\partial x}$ always positive: utility is always increasing in x
- $\frac{\partial U}{\partial x}$ increasing in y : the goods are *complements* in the sense that an increase in y raises the marginal value of x
- $\frac{\partial U}{\partial x}$ decreasing in x : x exhibits diminishing marginal utility

The rules of partial differentiation are identical to the rules of regular differentiation, treating all other variables as constants

Example: Let

$$U(w, x, y, z) = 10wxz + x^3 - wy + y^{\frac{1}{2}} - z^{10} + zw$$

Then:

$$\frac{\partial U}{\partial w} = 10xz - y + z$$

$$\frac{\partial U}{\partial x} = 10wz + 3x^2$$

$$\frac{\partial U}{\partial y} = -w + \frac{1}{2}y^{-\frac{1}{2}}$$

$$\frac{\partial U}{\partial z} = 10wx - 10z^9 + w$$

Second derivatives

The *second partial derivative* of f with respect to any variable x is written

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \frac{\partial f}{\partial x}$$

The *mixed partial derivatives* (or *cross partial derivatives*) of $f(x, y)$ are written

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \frac{\partial f}{\partial y}$$

Example:

$$U(w, x, y, z) = 10wxz + x^3 - wy + y^{\frac{1}{2}} - z^{10} + zw$$

$$\frac{\partial U}{\partial x} = 10wz + 3x^2$$

$$\frac{\partial^2 U}{\partial x^2} = 6x$$

$$\frac{\partial^2 U}{\partial x \partial w} = 10z$$

$$\frac{\partial^2 U}{\partial x \partial y} = 0$$

$$\frac{\partial^2 U}{\partial x \partial z} = 10w$$

etc...

Notation

There are many different ways to denote a partial derivative; the following are equivalent:

$$\frac{\partial f}{\partial x} \quad \frac{\partial}{\partial x} f \quad f_1 \text{ (provided } x \text{ is the first argument of } f) \quad f_x \quad D_x f$$

Similarly for cross partials the following are equivalent:

$$\frac{\partial^2 f}{\partial xy} \quad \frac{\partial}{\partial x} \frac{\partial f}{\partial y} \quad f_{12} \quad f_{xy} \quad D_{xy} f$$

The mixed derivative theorem

If f has continuous first, second and mixed partial derivatives (as in these derivatives exist and are themselves continuous), then f is a class C^2 function

A class C^k function is defined similarly, with k representing the number of partial derivatives f has that exist and are continuous (i.e. $\frac{\partial^k f}{\partial x_1 \dots}$)

Your book calls a class C^2 function *smooth*

Theorem: If f is a smooth function then $f_{xy} = f_{yx}$ for all x, y

Example:

$$U(w, x, y, z) = 10wxz + x^3 - wy + y^{\frac{1}{2}} - z^{10} + zw$$

$$\frac{\partial U}{\partial x} = 10wz + 3x^2$$

$$\frac{\partial U}{\partial y} = -w + \frac{1}{2}y^{-\frac{1}{2}}$$

$$\frac{\partial U}{\partial z} = 10wx - 10z^9 + w$$

$$\frac{\partial^2 U}{\partial x \partial y} = 0$$

$$\frac{\partial^2 U}{\partial y \partial x} = 0$$

$$\frac{\partial^2 U}{\partial x \partial z} = 10w$$

$$\frac{\partial^2 U}{\partial z \partial x} = 10w$$

etc...

The chain rule for two variables

Let $z = f(x, y)$ be a smooth function and suppose that x and y are themselves functions of a third variable, t

$$x = g(t) \text{ and } y = h(t)$$

Then

$$z = F(t) = f(g(t), h(t))$$

The *chain rule* says that

$$F'(t) = f_1(g(t), h(t))g'(t) + f_2(g(t), h(t))h'(t)$$

or alternatively,

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

Intuition: As t changes by a small amount, the overall change in z is the sum of the indirect effects of t 's change on x and y

Example: Let

$$f(x, y) = xy^4 + x^3y^2 \text{ and } x = 2 - 3t \text{ and } y = 4 + 5t$$

To find $f'(t)$ we could reexpress f in terms of t by substituting $x(t)$ and $y(t)$ back into f ; OR we could use the chain rule letting $z(t) = f(x, y)$

$$f(x, y) = xy^4 + x^3y^2 \text{ and } x = 2 - 3t \text{ and } y = 4 + 5t$$

$$\begin{aligned} \frac{dz}{dt} &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} \\ &= (y^4 + 3x^2y^2) \frac{dx}{dt} + (4xy^3 + 2x^3y) \frac{dy}{dt} \end{aligned}$$

$$\frac{dx}{dt} = -3$$

$$\frac{dy}{dt} = 5$$

$$\frac{dz}{dt} = -3(y^4 + 3x^2y^2) + 5(4xy^3 + 2x^3y)$$

You can convince yourself this is true by differentiating

$$(2 - 3t)(4 + 5t)^4 + (2 - 3t)^3(4 + 5t)^2$$

with respect to t and seeing if it equals

$$-3((4 + 5t)^4 + 3(2 - 3t)^2(4 + 5t)^2) + 5(4(2 - 3t)(4 + 5t)^3 + 2(2 - 3t)^3(4 + 5t))!$$

Total derivatives

Suppose that $y = h(x)$ and $z = F(x) = f(x, y)$; then $F'(x)$ is the *total derivative* of z with respect to x

This is different from $\frac{\partial f}{\partial x}$, which treats y as a constant; the total derivative accounts for the fact that a change in x causes both its own (direct) change in z , and also an indirect change in z through its effect on y

Using the chain rule we get that

$$\frac{dz}{dx} = f_x \frac{dx}{dx} + f_y \frac{dy}{dx}$$

$$\frac{dz}{dx} = \underbrace{f_x}_{\text{Direct effect of } x \text{ on } f} + \underbrace{f_y dy/dx}_{\text{Indirect effect of } x \text{ on } f \text{ via } y}$$

Suppose that $f(x_1, x_2, \dots, x_k)$ is a smooth function of k variables, where each x_k is itself a function of some other variable, t

$$f(x_1, x_2, \dots, x_k) = F(t)$$

The chain rule says that

$$\frac{dF}{dt} = \sum_{i=1}^k \frac{\partial f}{\partial x_i} \frac{dx_i}{dt}$$

Now suppose that each x_k is a function of *several* other variables, $t, u, \dots$

$$f(x_1, x_2, \dots, x_k) = F(t, u, \dots)$$

The chain rule says that

$$\frac{\partial F}{\partial t} = \sum_{i=1}^k \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial t}$$

The gradient

Let $f(x_1, x_2, \dots, x_k)$ be a function of k variables; at each point $(x_1, x_2, \dots, x_k)$ we define the gradient vector $Df(x_1, \dots, x_k)$ as follows:

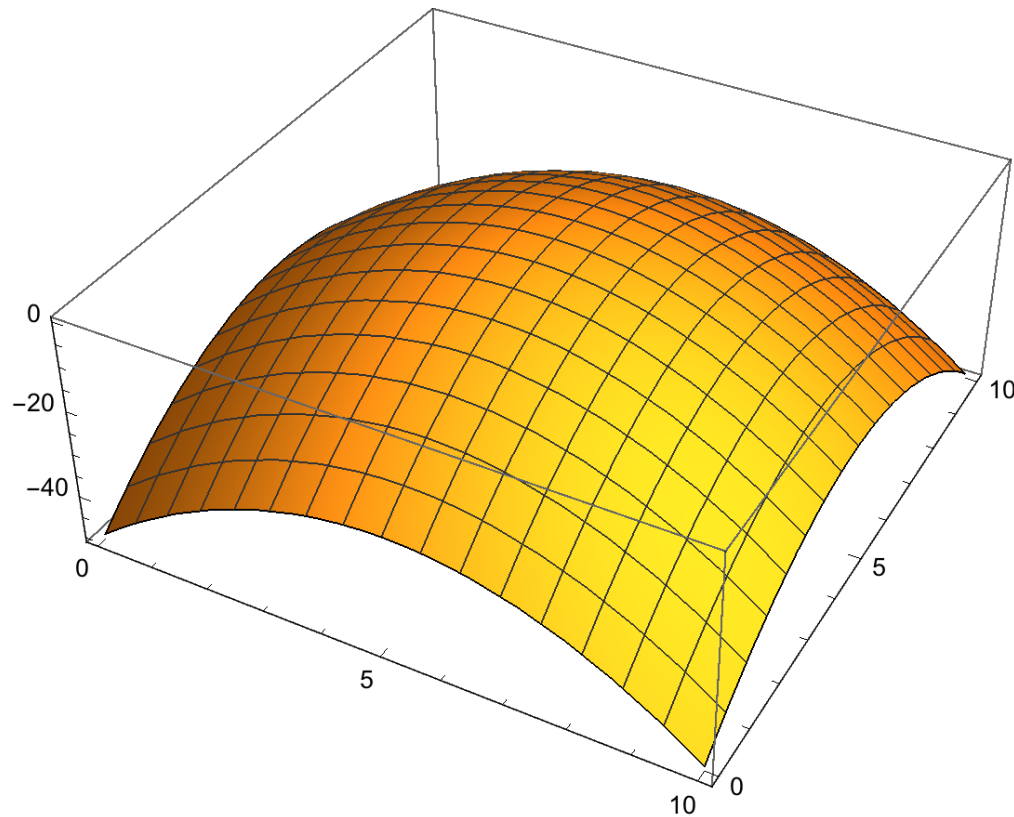
$$Df(x_1, \dots, x_k) = \nabla f(x_1, \dots, x_k) = \begin{bmatrix} f_{x_1}(x_1, x_2, \dots, x_k) \\ f_{x_2}(x_1, x_2, \dots, x_k) \\ \dots \\ f_{x_k}(x_1, x_2, \dots, x_k) \end{bmatrix}$$

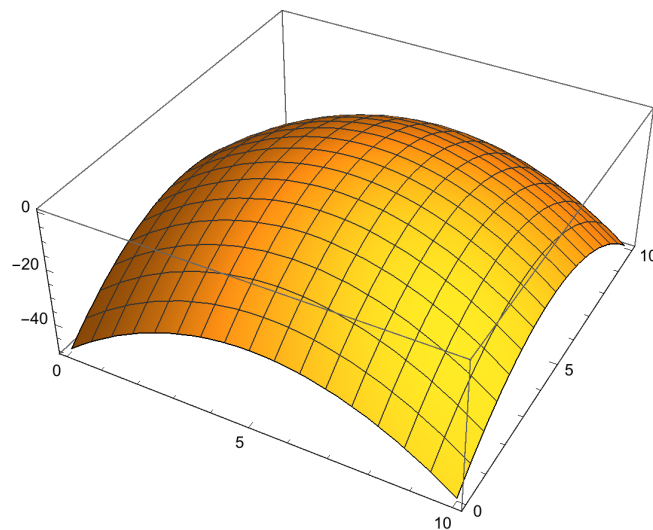
It's the vector whose components are the k partial derivatives of f

$\nabla f(\mathbf{x})$ points in the direction of the greatest rate of increase of f evaluated at $\mathbf{x}$; its magnitude is the slope of f in that direction

Example:

$$U(x, y) = -(x - 5)^2 - (y - 5)^2$$





$$\nabla U(x, y) = (10 - 2x, 10 - 2y)$$

$$\nabla U(0, 0) = (10, 10)$$

$$\nabla U(2, 2) = (6, 6)$$

$$\nabla U(0, 5) = (10, 0)$$

$$\nabla U(0, 2.5) = (10, 5)$$

The Hessian matrix

A matrix that consists of all of the cross-partial derivatives of $f(x_1, x_2, \dots, x_k) = f(\mathbf{x})$ in the following way:

$$D^2 f(\mathbf{x}) = \begin{bmatrix} f_{x_1 x_1}(\mathbf{x}) & f_{x_1 x_2}(\mathbf{x}) & \dots & f_{x_1 x_k}(\mathbf{x}) \\ f_{x_2 x_1}(\mathbf{x}) & f_{x_2 x_2}(\mathbf{x}) & \dots & f_{x_2 x_k}(\mathbf{x}) \\ \dots & \dots & \dots & \dots \\ f_{x_k x_1}(\mathbf{x}) & f_{x_k x_2}(\mathbf{x}) & \dots & f_{x_k x_k}(\mathbf{x}) \end{bmatrix}$$

This matrix is square and symmetric (why?)

Example

$$U(x, y) = -(x - 5)^2 - (y - 5)^2$$

$$\nabla U(x, y) = (10 - 2x, 10 - 2y)$$

$$D^2 f(\mathbf{x}) = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$$

Critical points and their classification

For a function of many variables, $f(x_1, x_2, \dots, x_k) = f(\mathbf{x})$, critical points $\mathbf{x}$ are defined as points where the gradient vector $\nabla f(\mathbf{x})$ equals zero

As with functions of one variable, we use second-order conditions to determine what kind of critical point we have:

- 1 If f has a local maximum at $\mathbf{x}^*$ then $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $D^2 f(\mathbf{x}^*)$ is negative semidefinite
- 2 If $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $D^2 f(\mathbf{x}^*)$ is negative definite then f has a local maximum at $\mathbf{x}^*$
- 1' If f has a local minimum at $\mathbf{x}^*$ then $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $D^2 f(\mathbf{x}^*)$ is positive semidefinite
- 2' If $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $D^2 f(\mathbf{x}^*)$ is positive definite then f has a local minimum at $\mathbf{x}^*$

Example

$$U(x, y) = -(x - 5)^2 - (y - 5)^2$$

$$\nabla U(x, y) = (10 - 2x, 10 - 2y)$$

$$D^2 f(\mathbf{x}) = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$$

The Hessian is negative definite for all $\mathbf{x}$ because the determinant is always strictly positive and both diagonal entries are strictly negative

Critical point $\mathbf{x} = (x, y) = (5, 5)$ is a local maximum

Global optima

Knowing that $\mathbf{x}$ is a local maximum doesn't help us if we are looking for a global maximum; if f is concave or convex then we know a lot more

$f : \mathbb{R}^n \rightarrow \mathbb{R}$ is *concave* if and only if for any $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$ and any $\alpha \in [0, 1]$:

$$f(\alpha \mathbf{x}_1 + (1 - \alpha) \mathbf{x}_2) \geq \alpha f(\mathbf{x}_1) + (1 - \alpha) f(\mathbf{x}_2)$$

- The function $f(\mathbf{x})$ is concave (convex) if and only if the Hessian $D^2 f(\mathbf{x})$ is negative (positive) semidefinite FOR ALL $\mathbf{x}$
- The concave (convex) function $f(\mathbf{x})$ attains a global maximum (minimum) at $\mathbf{x}^*$ if and only if the gradient $\nabla f(\mathbf{x}^*) = \mathbf{0}$

$\Rightarrow$ Every critical point of a concave function is a global maximum

$\Rightarrow$ Every critical point of a convex function is a global minimum

Concavity / convexity is tremendously helpful in classifying critical points; the following are tricks that can be used to help determine whether a function is concave / convex:

- The sum of two concave functions is concave; the sum of two convex functions is convex
- A concave function of one variable is a concave function of many variables: if $f(x_1, x_2, \dots, x_k) = g(x_1)$ with g concave, then f is concave
- f is concave if and only if $-f$ is convex
- A *strictly* concave function has at most one global maximum
- Let g be a nondecreasing and concave function of one variable; then if $f(\mathbf{x})$ is concave so is $g(f(\mathbf{x}))$
- Let h be a nondecreasing and convex function of one variable; then if $f(\mathbf{x})$ is convex so is $h(f(\mathbf{x}))$

Example: Let

$$U(x, y) = -(x - 2)(y + 5)$$

Find the critical points of U

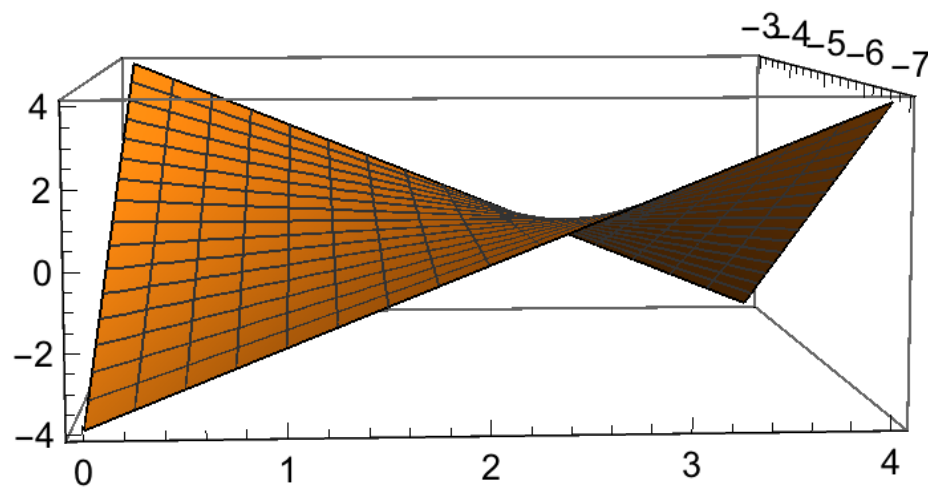
Can you classify the critical point(s)?

$$U(x, y) = -(x - 2)(y + 5)$$

Critical point: $(2, -5)$

$$D^2U = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

$\text{Det}[D^2U] = -1 < 0$, and so the critical point is a saddle point



Example: Let

$$U(x, y) = -x^2 + 4xy - 4y^2$$

Find the critical points of U

Can you classify the critical point(s)?

$$U(x, y) = -x^2 + 4xy - 4y^2$$

An infinite number of critical points: $(2C, C)$ for all $C \in \mathbb{R}$

$$D^2U = \begin{bmatrix} -2 & 4 \\ 4 & -8 \end{bmatrix}$$

$\text{Det}[D^2U] = 0$, and all diagonals are negative, so Hessian is negative semidefinite for all $\mathbf{x}$ and U is concave: all critical points are global optima

